

Daniel Gomm

PhD Candidate in Computer Science

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I work at the intersection of generative AI, information retrieval, and structured data. My research targets *open tabular insight extraction*: enabling anyone to get reliable answers from large, heterogeneous collections of tables such as data lakes. My recent work frames this problem end to end, characterizes ambiguity in natural-language inputs, and improves table retrieval.

Education

2024.10 – now Amsterdam, NL	PhD in Computer Science Centrum Wiskunde & Informatica – University of Amsterdam Contextualizing generative AI in structured data. Advisors: Madelon Hulsebos and Maarten de Rijke. – Developed the perspective of <i>Open Tabular Insight Extraction</i>: a unified problem formulation that harmonizes table QA, text-to-SQL, table retrieval, and data-analysis agents, backed by a systematic review across ML, IR, NLP, DB, and HCI literature (under review, preprint available). – Contributions to interpreting ambiguity in natural language inputs, more robust and performant table retrieval, improving text-to-SQL generation, and agentic data science.
2020.10 – 2023.12 Karlsruhe, DE	M.Sc. Industrial Engineering Karlsruhe Institute of Technology (KIT) Final grade 1.0 (German scale, 1.0 = best). Focus: data engineering and machine learning. Thesis: <i>Counterfactual Explanations for Black-Box Models on Dynamic Graphs</i> – basis of ICML’25 paper.
2022.01 – 2022.06 Espoo, FI	Exchange Semester, M.Sc. Computer Science Aalto University Grade 4.7 / 5.0. Focus: big-data platforms and deep learning.
2016.10 – 2020.10 Karlsruhe, DE	B.Sc. Industrial Engineering Karlsruhe Institute of Technology (KIT) Final grade 1.7. Thesis: <i>Adapting Stream Processing Pipelines in Fog Infrastructures</i> .

Research & Industry Experience

2024.04 – 2024.08 Ispra, IT	Scientific Trainee European Commission, Joint Research Centre (JRC) – European Centre for Algorithmic Transparency – Researched RAG and agentic approaches for auditing algorithmic systems to support enforcement of the EU Digital Services Act.
2022.10 – 2023.03 Munich, DE	Working Student, Software Engineering Intel – Developed features for System Trace, Intel’s tool for debugging and tracing pre-production silicon.
2020.11 – 2021.12 Karlsruhe, DE	Research Assistant FZI Research Center for Information Technology – Integrated dynamic edge–cloud workload migration (from my B.Sc. thesis) into Apache StreamPipes, an industrial-IoT stream-processing toolbox.
2019.10 – 2020.03 Stuttgart, DE	Intern, IT Program Management Robert Bosch GmbH – On my own initiative, built a classification model for part numbers to automate a manual workflow previously handled by a multi-person team.

Publications

Conference Papers

ICML 2025 **CoDy: Counterfactual Explainers for Dynamic Graphs**
Zhan Qu^{*}, **Daniel Gomm**^{*}, Michael Färber
International Conference on Machine Learning (ICML), PMLR 267, 2025 [Paper](#) → [Code](#) →

Workshop Papers

EurIPS 2025 **Are We Asking the Right Questions? On Ambiguity in Natural Language Queries for Tabular Data Analysis**
Daniel Gomm, Cornelius Wolff, Madelon Hulsebos
AI for Tabular Data Workshop @ EurIPS, 2025 – **Spotlight** [Paper](#) → [Talk](#) →

EurIPS 2025 **SQaLe: A Large Text-to-SQL Corpus Grounded in Real Schemas**
Cornelius Wolff, **Daniel Gomm**, Madelon Hulsebos
AI for Tabular Data Workshop @ EurIPS, 2025 [Paper](#) → [Data](#) →

ELLIS 2025 **Metadata Matters in Dense Table Retrieval**
Daniel Gomm, Madelon Hulsebos
ELLIS Workshop on Representation Learning and Generative Models for Structured Data, 2025 [Paper](#) →

Preprints

arXiv 2026 **Open Tabular Insight Extraction: Where Do We Stand, and Where Should We Go?**
Daniel Gomm, Maarten de Rijke, Madelon Hulsebos
arXiv preprint, 2026 [arXiv](#) →

Preprint 2025 **Unlocking the Full Potential of Data Science Requires Tabular Foundation Models, Agents, and Humans**
Tianji Cong^{*}, Julian Martin Eisenschlos^{*}, **Daniel Gomm**^{*}, Leo Grinsztajn^{*}, Andreas C. Mueller^{*}, Anupam Sanghi^{*}, et al. (24 authors)
Position paper, 2025 [Paper](#) →

^{*} Equal contribution / shared first authorship.

Open-Source Packages & Tools

PyPI
openbatch **OpenBatch**
Typed Python interface for building OpenAI Batch API jobs; drop-in replacement for the openai client. [Code](#) → [Blog](#) →

PyPI
tableserializer **Table Serialization Kitchen**
Composable table-to-text serialization for LLMs (QA, text-to-SQL, retrieval) with reproducible experiments through the TARGET benchmark. [Code](#) → [Blog](#) →

PyPI
pytei-client **PyTEI**
Minimal Python client for Hugging Face Text Embeddings Inference with persistent caching. [Code](#) →

Talks

2026.01 *Are We Asking the Right Questions?* – TRL Seminar (invited)

2025.12 *Are We Asking the Right Questions?* – Spotlight, AI for Tabular Data Workshop @ EurIPS 2025

Teaching & Supervision

M.Sc. Thesis Supervision

2025.03 – 2025.10 **Nikola Selic** · M.Sc. Computer Science, TU Munich

2026.02 – now	Yme Kingma · M.Sc. Data Science, Radboud University Thesis: <i>Exploring Unanswerability in Table Question Answering – From Cause to Detection</i>
Teaching	
2018.10 – 2019.08 Karlsruhe, DE	Teaching Assistant, Java Programming Karlsruhe Institute of Technology (KIT) Theoretical and practical tutorials for introductory and advanced Java courses.

Academic Service

Reviewing	AI for Tabular Data Workshop @ EurIPS 2025; KDD 2026 (external reviewer)
Community	Founder and organizer, Amsterdam Lunch on Table (ALOT) reading group, since 2025
Volunteering	DBDBD 2024, CIDR 2025, ELLIS Workshop 2025

Skills

Programming	Python, SQL, Java, TypeScript/JavaScript, Bash
LLMs & ML	PyTorch, Hugging Face (Transformers, Datasets), vLLM, llama.cpp, LLM APIs, NumPy, pandas
Infrastructure	Linux, Docker, Slurm, Kubernetes, Git
Languages	German (native), English (fluent), Spanish (B1), Dutch (A2)